

FINAL PRESENTATION · NLP COURSE

Clinical Infection Risk Assessment

Flagging patients at high infection risk from free-text clinical notes - trained on synthetic data generated from real MIMIC-IV records.

PRESENTED BY

Tal Meillet & Hodaya Yasayev Klenter

COURSE INSTRUCTOR

Sasha Apartsin

Reading infection risk from the clinical narrative

~136M

healthcare-associated infections
worldwide each year

+13days

typical added length of stay per
infection

~\$8k

average added cost per infection-driven
readmission - borne by hospitals and
patients alike

Source: AHRQ/NIS analysis, 91,613 SSI readmissions ≈ \$700M/yr
(PubMed 19398246)

THE PROBLEM

Early signs of infection appear in **free-text clinical notes** , but fixed scores like SOFA / qSOFA read only structured numbers and **miss the narrative** .

THE CHALLENGE

Labeled notes are **locked behind privacy law** and costly to annotate. The **data - not the model** - is the real bottleneck.

WHAT'S NOVEL

A problem with no data

No dataset existed - so we **built the corpus itself** .

Fact-then-write, anti-leakage

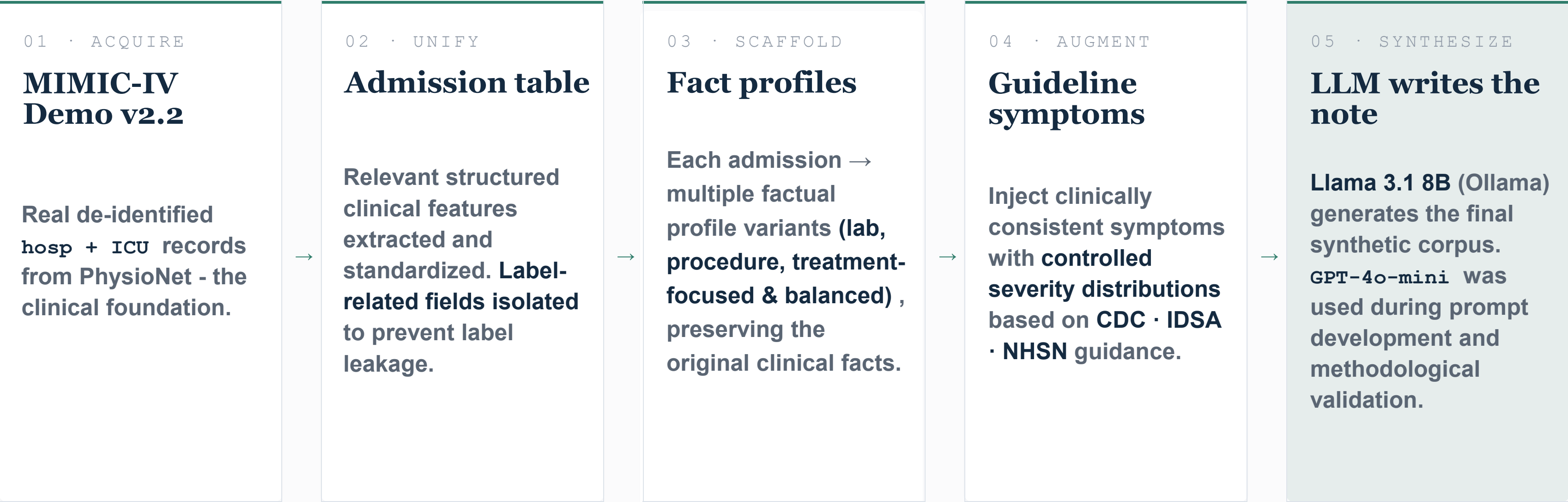
Facts fixed by rules; the LLM only **phrases** them - and giveaway terms are generalized.

THE TASK

Classify a clinical note by infection concern - a binary decision-support signal.

note → **High** / **Low**

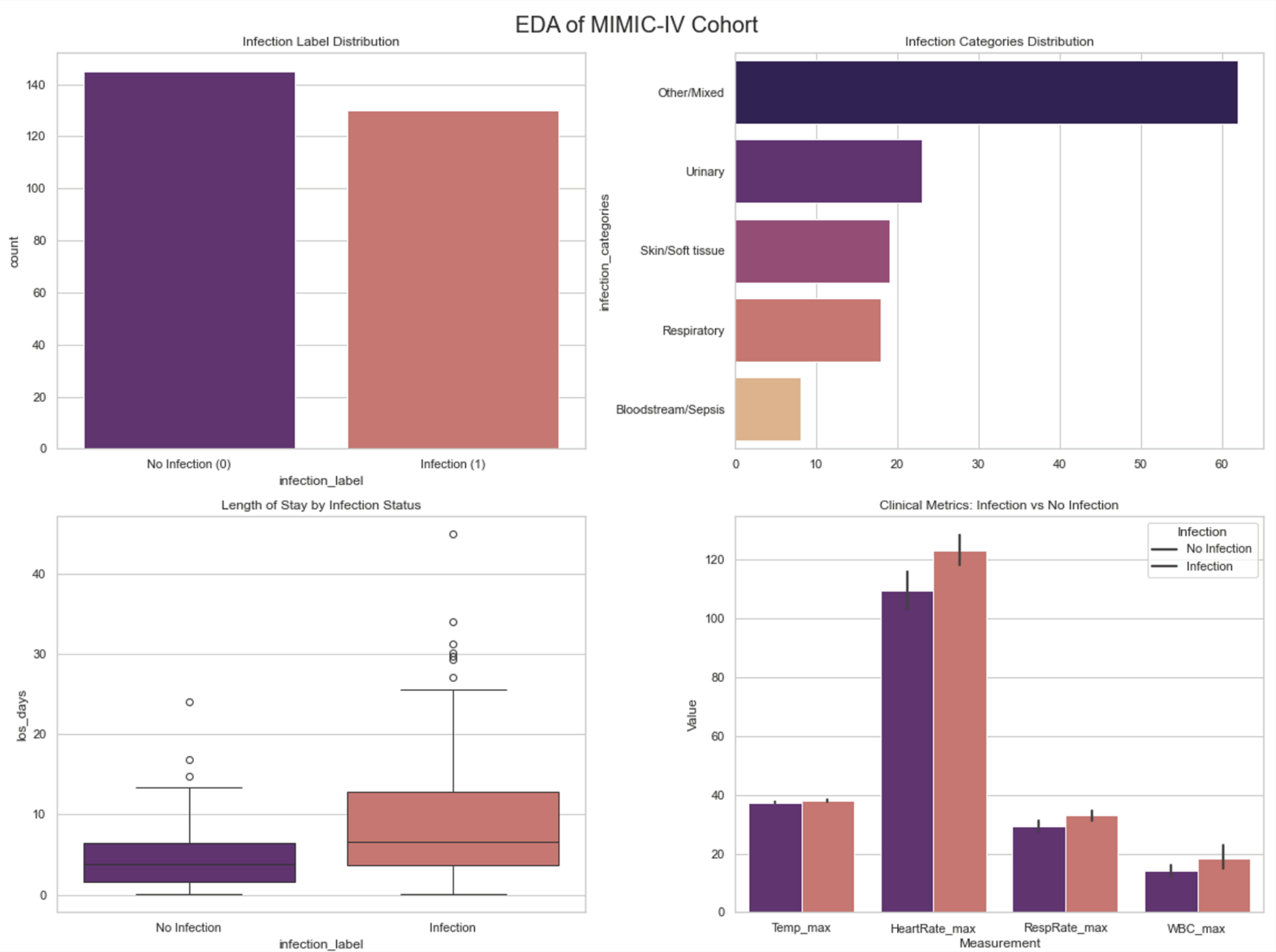
From structured records to realistic clinical notes



GUIDING PRINCIPLE

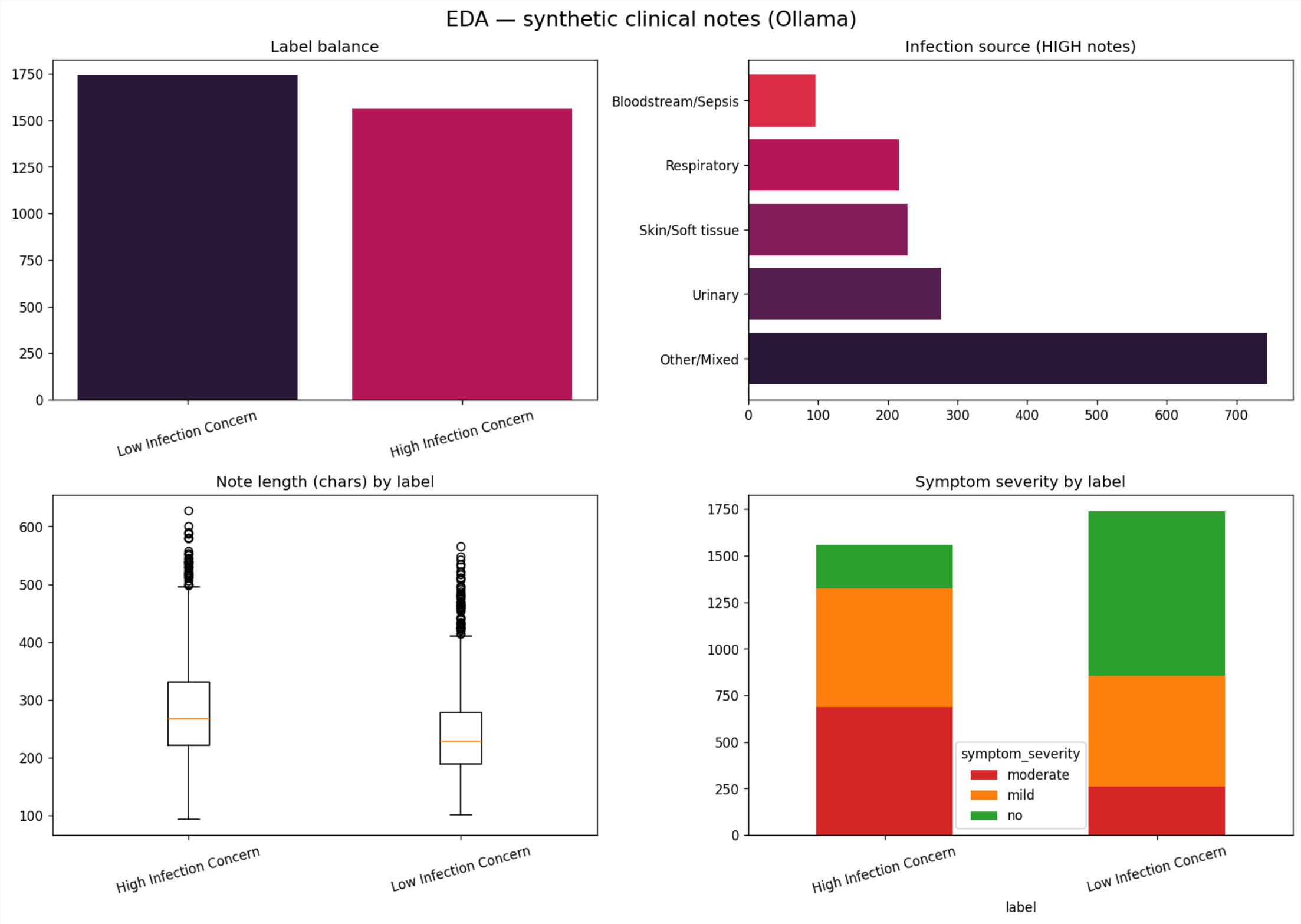
The model is constrained to generate notes only from pre-verified structured facts , minimizing hallucinations and label leakage.

The source cohort - 274 MIMIC-IV admissions



From the main visualization:
Infection labels are **near-balanced** across the 274 admissions, so the model sees both classes in roughly equal measure during training. Infected patients consistently run **higher heart rate, respiratory rate & WBC**, and stay longer - exactly the physiological signal the clinical notes need to carry through to the text.

The synthetic corpus - ~3,300 generated notes



Label balance, infection sources, and note length **closely mirror the real MIMIC-IV cohort** , even though every note was generated. Severity of symptoms also tracks the assigned label consistently - evidence that the **fact-then-write pipeline stayed faithful** to the underlying clinical facts rather than drifting into generic template language.

Model Evaluation: Zero-shot Baseline vs. Fine-tuned Classifiers

THREE MODELS, ONE TASK

Zero-shot Llama 3.1 (Ollama) - the baseline

The same local **Llama 3.1** used to generate the corpus, prompted directly for infection concern with **no fine-tuning**. Sets the bar a trained classifier has to clear. **Measures how well an LLM performs without seeing our training data.**

GPT-2 - the fine-tuned general model

A general-purpose language model with a classification head and **no medical pretraining** - it has to learn clinical vocabulary from our corpus alone.

PubMedBERT - the domain model

A BERT encoder **pretrained on millions of PubMed articles**, so it already captures biomedical language patterns before we fine-tune it on our synthetic clinical notes. [microsoft/BiomedNL-PubMedBERT](#)

WHY ACCURACY ALONE IS NOT ENOUGH

Plain accuracy misleads here: the two errors carry **very different clinical cost**, and a model can score well by just predicting the common class. so we track four:

Recall (High)

Of truly concerning notes, the share we catch. **Our top priority** - a missed High is a real risk left unflagged.

Precision (High)

Of the notes we flag, the share that are right. Controls **false alarms and alert fatigue**.

F1 & ROC-AUC

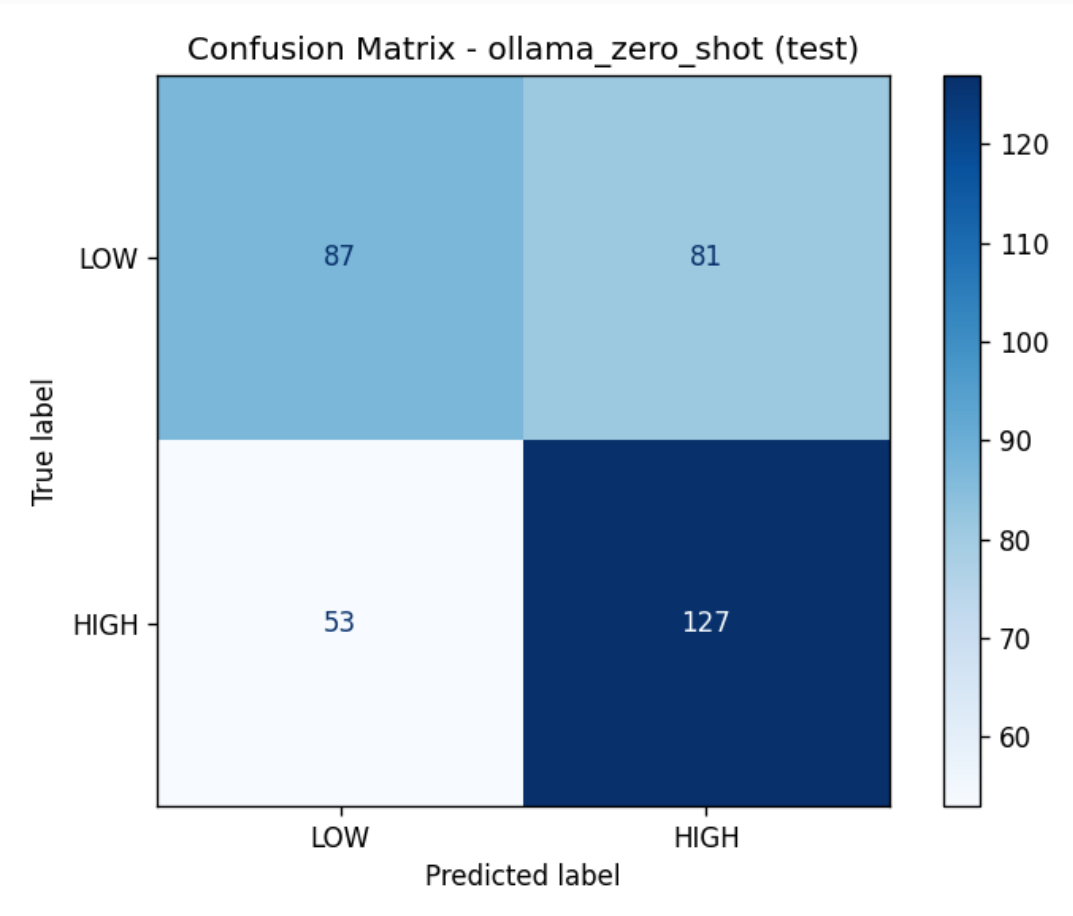
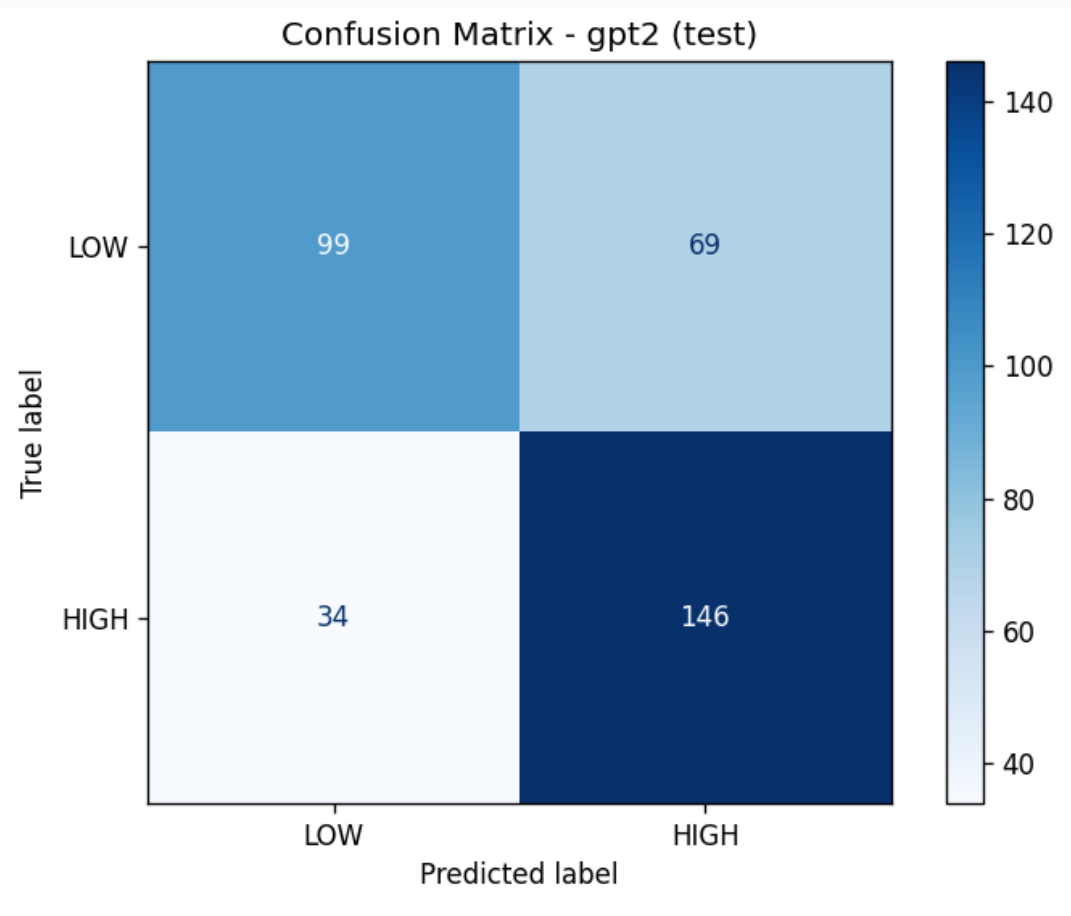
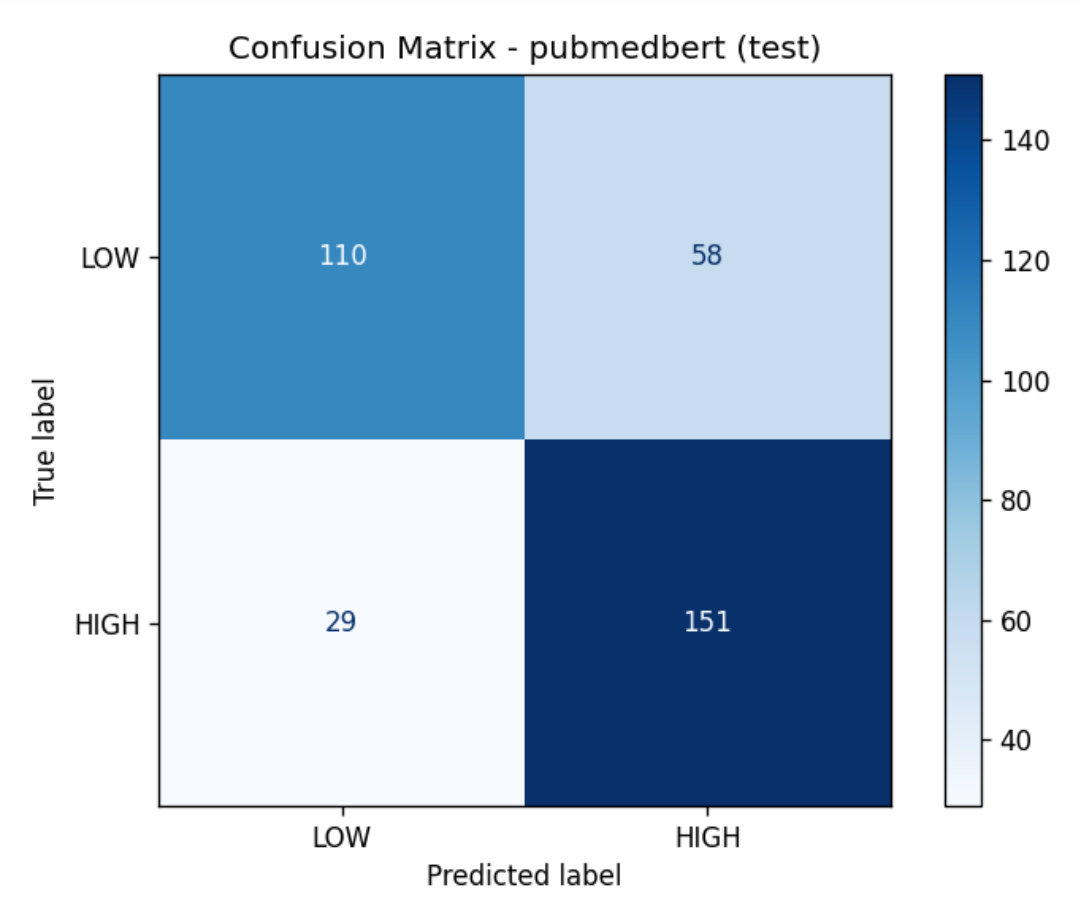
F1 balances recall and precision into one comparable number; ROC-AUC rates ranking quality **across all thresholds**, independent of the cutoff.

PubMedBERT wins on every metric

MODEL	RECALL (HIGH)	PRECISION (HIGH)	F1 (HIGH)	ROC-AUC
PubMedBERT biomedical encoder	0.839	0.722	0.776	0.812
GPT-2 general-purpose model	0.811	0.679	0.739	0.746
Zero-shot Llama 3.1 baseline · no fine-tuning	0.706	0.610	0.655	—

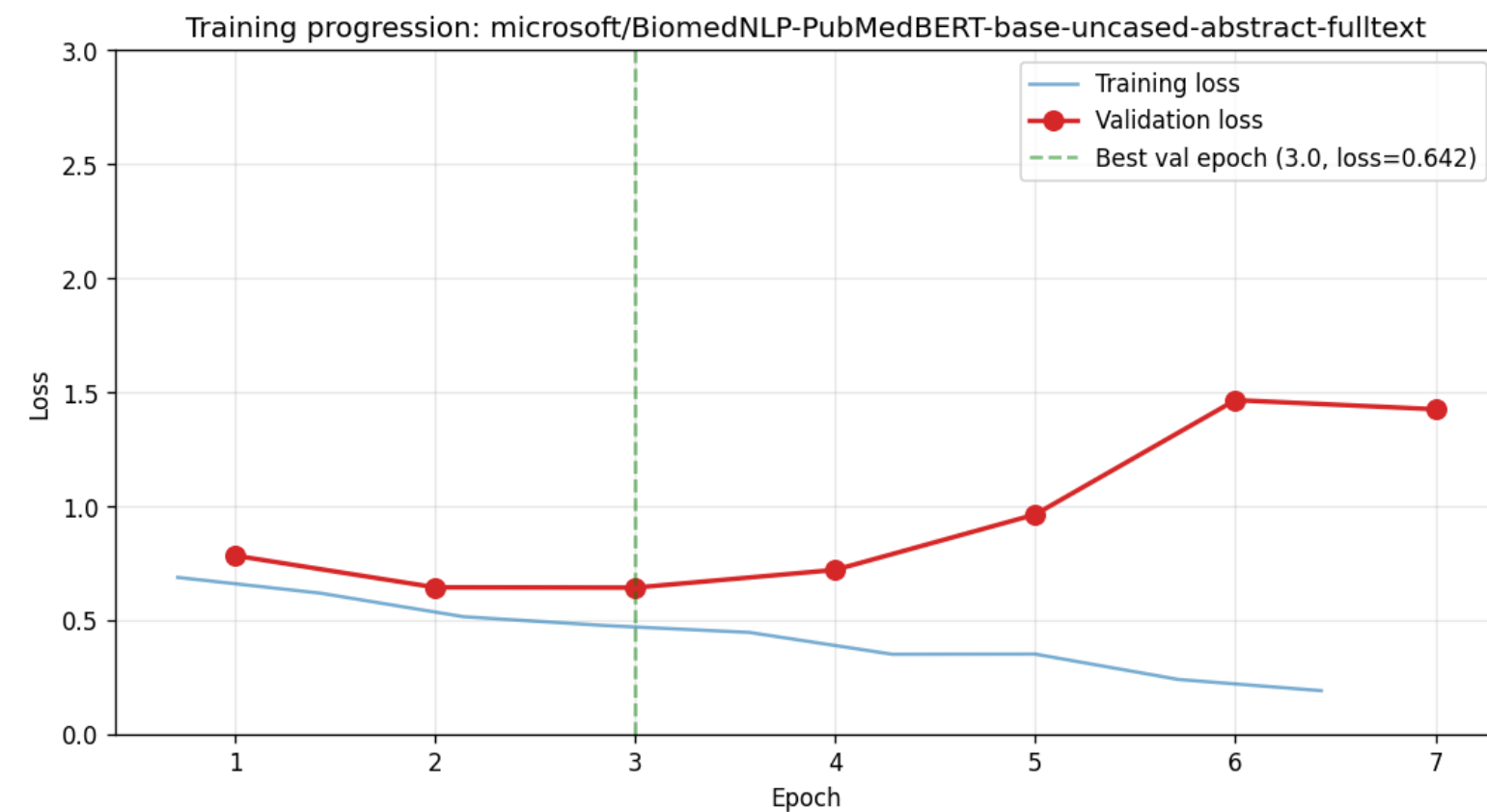
Both fine-tuned classifiers clear the zero-shot bar comfortably; **PubMedBERT leads on every metric** , catching ~84% of true high-concern notes at ~72% precision-usable as a triage aid with a human in the loop.

Zero-shot has no ROC-AUC: it only ever outputs a label, not a probability score, so there's no threshold curve to compute AUC from.



How the fine-tuning models learned

TRAINING PROGRESSION · PUBMEDBERT



TRAINING PROGRESSION · GPT-2



Each model is checkpointed at its **best validation epoch** (dashed line) and that snapshot is what we evaluate on the test set. Training loss keeps falling while validation loss turns back up — the classic sign of **overfitting** past the optimum, which early checkpointing avoids.

What we learned & where it goes next

CONCLUSIONS

The pipeline works end-to-end

From raw MIMIC-IV admissions to a trained infection-concern classifier — the full scaffold-and-augment approach produced a usable, labeled corpus where none existed.

Domain pretraining paid off

Fine-tuning substantially outperformed the zero-shot baseline, while PubMedBERT achieved the best overall performance across all evaluation metrics, supporting the use of a biomedically pretrained encoder for clinical text classification.

Anti-leakage held up

The same clinically plausible error pattern appeared across all three models : Bloodstream/Sepsis cases were consistently easiest, while Skin/Soft tissue, Urinary, and true LOW-concern cases remained the most challenging — suggesting the models learned genuine clinical difficulty rather than exploiting label shortcuts.

Good, not yet clinical

Fine-tuning substantially improved performance over direct zero-shot prompting, but the models should still be viewed as clinical decision-support tools rather than autonomous decision-makers.

FUTURE WORK

- Refine note-generation prompts for **skin / soft-tissue** cases.
- Explore ensembles of the fine-tuned models and targeted augmentation of the most challenging infection categories (especially **Skin/Soft tissue and Urinary** cases).
- Scale beyond the demo to full MIMIC-IV.

Caveat: demo cohort (~100 patients) — a proof-of-concept; re-validate at scale before any clinical claim.

Zero-shot gap: the largest gain from fine-tuning was on **Urinary infections** , where the zero-shot baseline performed substantially worse than both fine-tuned models.